

Computer Communication & Networks

Lecture 19

Interconnecting LANs

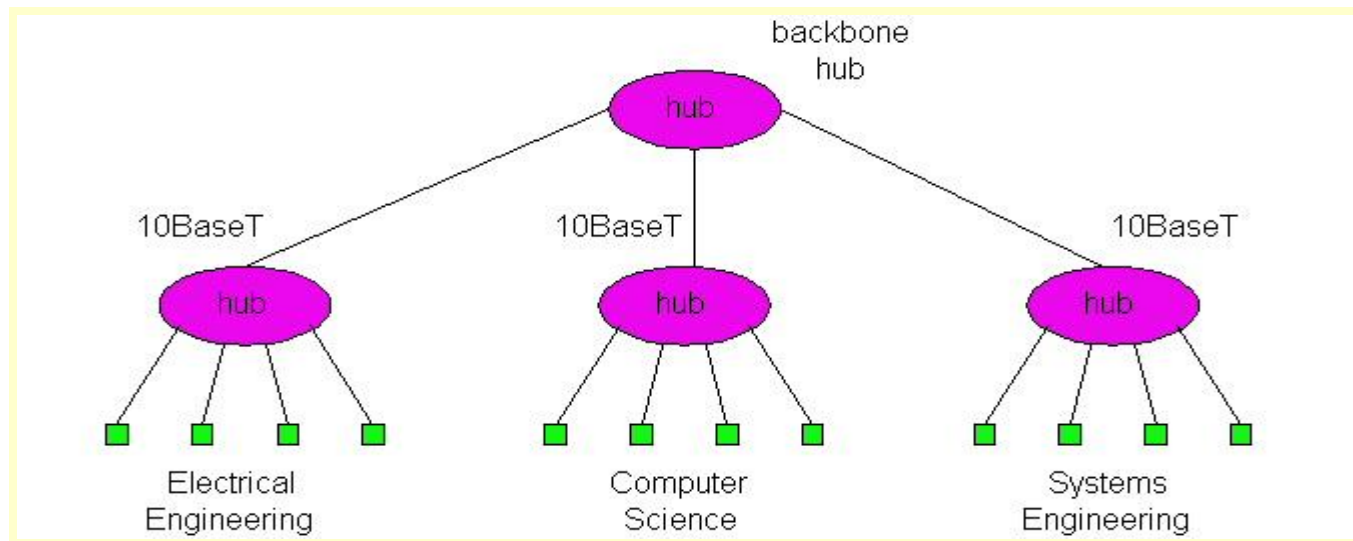
Interconnecting LANs

Q: Why not just one big LAN?

- **Limited amount of supportable traffic: on single LAN, all stations must share bandwidth**
- **Limited length: 802.3 specifies maximum cable length**
- **Large “collision domain” (can collide with many stations)**

Hubs - 1

- **Physical Layer devices: essentially repeaters operating at bit levels: repeat received bits on one interface to all other interfaces**
- **Hubs can be arranged in a **hierarchy** (or multi-tier design), with **backbone** hub at its top**



Hubs - 2

- Each connected LAN referred to as LAN segment
- Hubs **do not isolate** collision domains: node may collide with any node residing at any segment in LAN
- Hub Advantages:
 - simple, inexpensive device
 - Multi-tier provides graceful degradation: portions of the LAN continue to operate if one hub malfunctions
 - extends maximum distance between node pairs (100m per Hub)

Hub limitations

- **Single collision domain results in no increase in max throughput**
 - **multi-tier throughput same as single segment throughput**
- **Individual LAN restrictions pose limits on number of nodes in same collision domain and on total allowed geographical coverage**
- **Cannot connect different Ethernet types (e.g., 10BaseT and 100baseT)**

Bridges - 1

- **Link Layer devices:** operate on Ethernet frames, examining frame header and selectively forwarding frame based on its destination
- Bridge **isolates collision** domains since it buffers frames
- When frame is to be forwarded on segment, bridge uses CSMA/CD to access segment and transmit

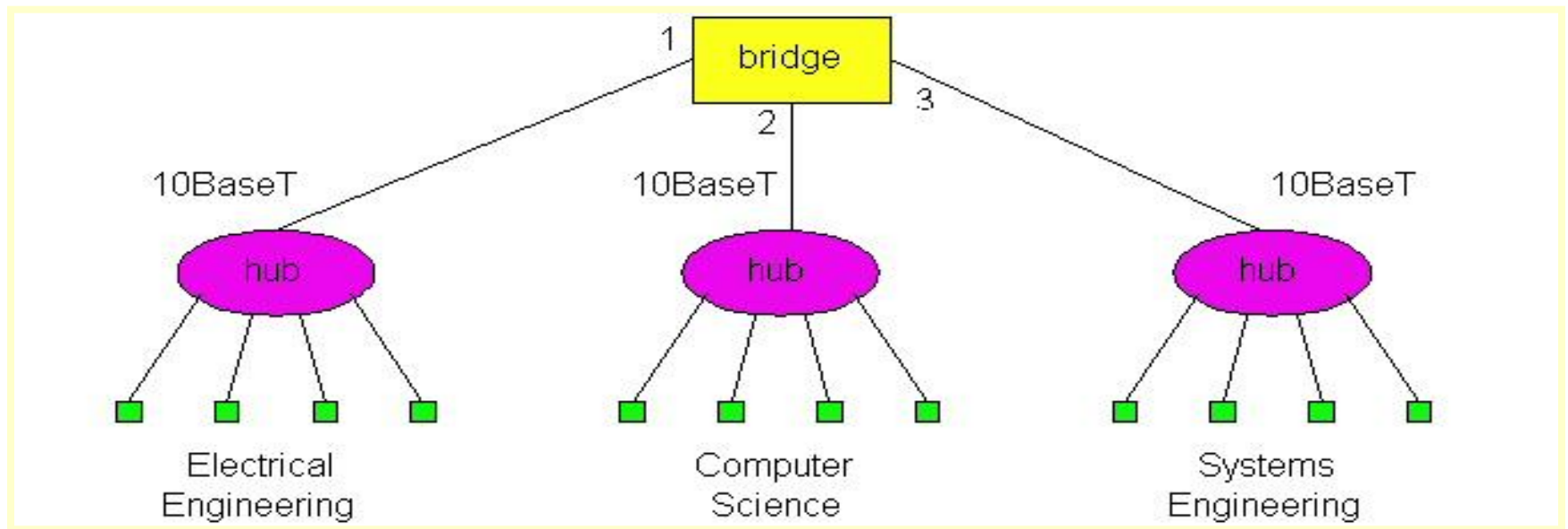
Bridges - 2

- **Bridge advantages:**
 - ❑ **Isolates collision domains resulting in higher total max throughput, and does not limit the number of nodes nor geographical coverage**
 - ❑ **Can connect different type Ethernet since it is a store and forward device**
 - ❑ **Transparent: no need for any change to hosts LAN adapters**

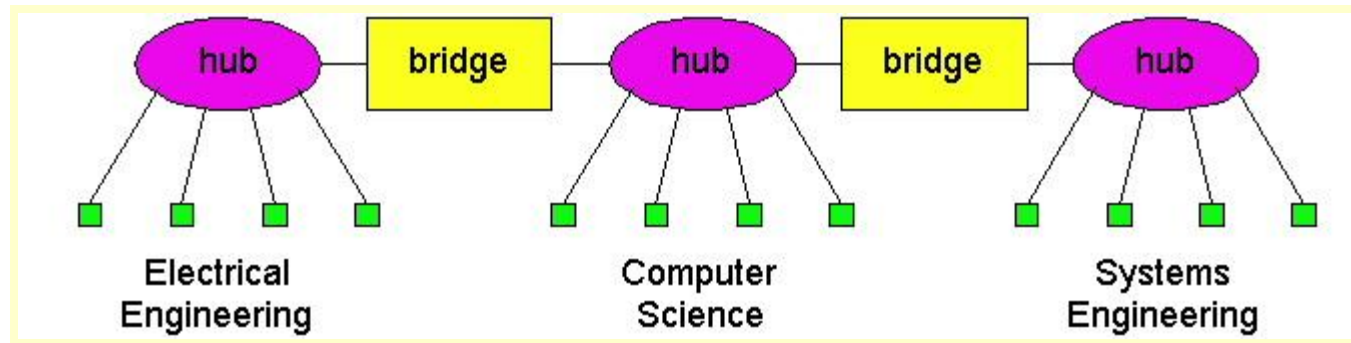
Bridges: frame filtering, forwarding

- **Bridges filter packets**
 - **Same-LAN -segment frames not forwarded onto other LAN segments**
- **Forwarding:**
 - **How to know which LAN segment on which to forward frame?**
 - **Looks like a routing problem (more shortly!)**

Backbone Bridge



Interconnection Without Backbone



- **Not recommended for two reasons:**
 - **Single point of failure at Computer Science hub**
 - **All traffic between EE and SE must path over CS segment**

Bridge Filtering - 1

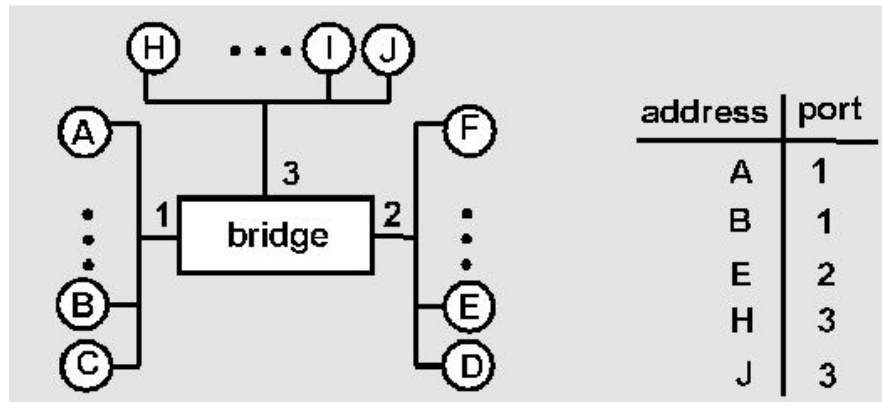
- Bridges **learn** which hosts can be reached through which interfaces: maintain filtering tables
 - when frame received, bridge “learns” location of sender: incoming LAN segment
 - records sender location in filtering table
- Filtering table entry:
 - (Node LAN Address, Bridge Interface, Time Stamp)
 - stale entries in Filtering Table dropped (TTL can be 60 minutes)

Bridge Filtering - 2

- Filtering procedure:
 - if** destination is on LAN on which frame was received
 - then** drop the frame
 - else** { lookup filtering table
 - if** entry found for destination
 - then** forward the frame on interface indicated;
 - else** flood; */* forward on all but the interface on which the frame arrived*/*
 - }

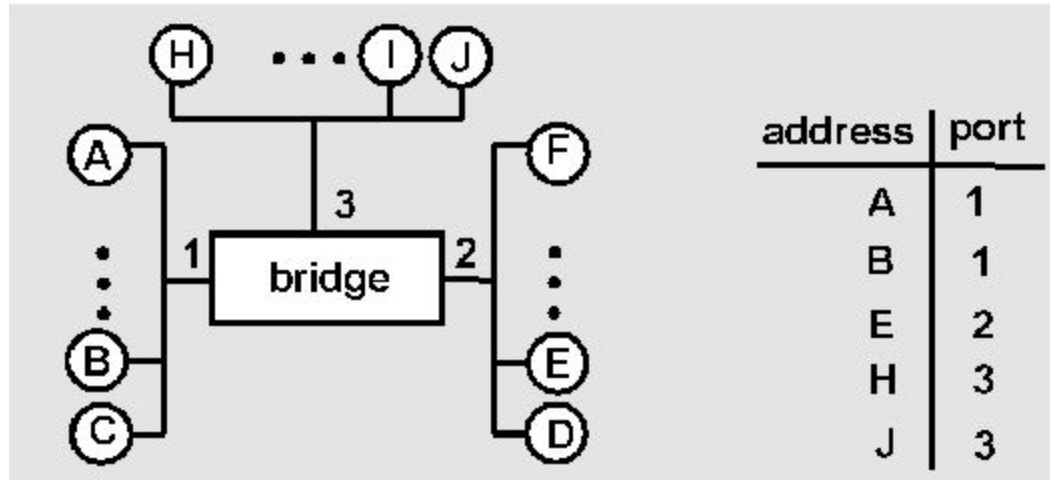
Bridge Learning: example - 1

Suppose C sends frame to D and D replies back with frame to C



- C sends frame, bridge has no info about D, so floods to both LANs
 - ❑ bridge notes that C is on port 1
 - ❑ frame ignored on upper LAN
 - ❑ frame received by D

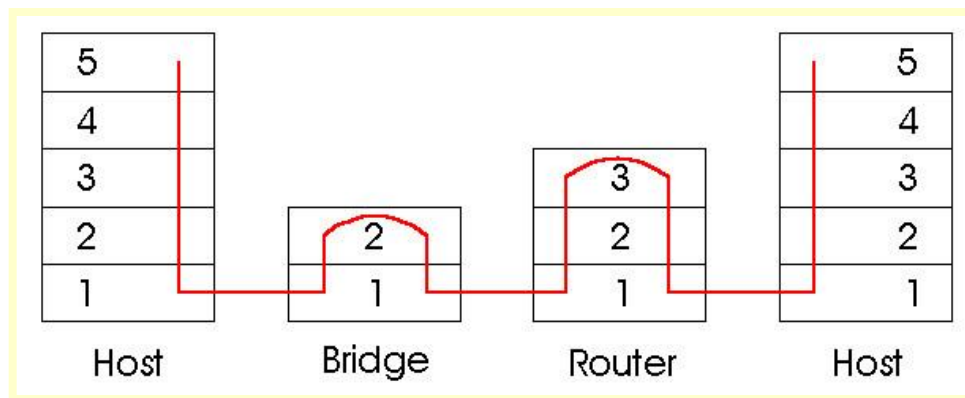
Bridge Learning: example - 2



- D generates reply to C, sends
 - ❑ bridge sees frame from D
 - ❑ bridge notes that D is on interface 2
 - ❑ bridge knows C on interface 1, so *selectively* forwards frame out via interface 1

WWF Bridges vs. Routers

- **Both store-and-forward devices**
 - ❑ **routers: network layer devices (examine network layer headers)**
 - ❑ **bridges are Link Layer devices**
- **Routers maintain routing tables, implement routing algorithms**
- **Bridges maintain filtering tables, implement filtering, learning and spanning tree algorithms**



Routers vs. Bridges - 1

Bridges + and -

- + Bridge operation is simpler requiring less processing bandwidth**
- Topologies are restricted with bridges: a spanning tree must be built to avoid cycles**
- Bridges do not offer protection from broadcast storms (endless broadcasting by a host will be forwarded by a bridge)**

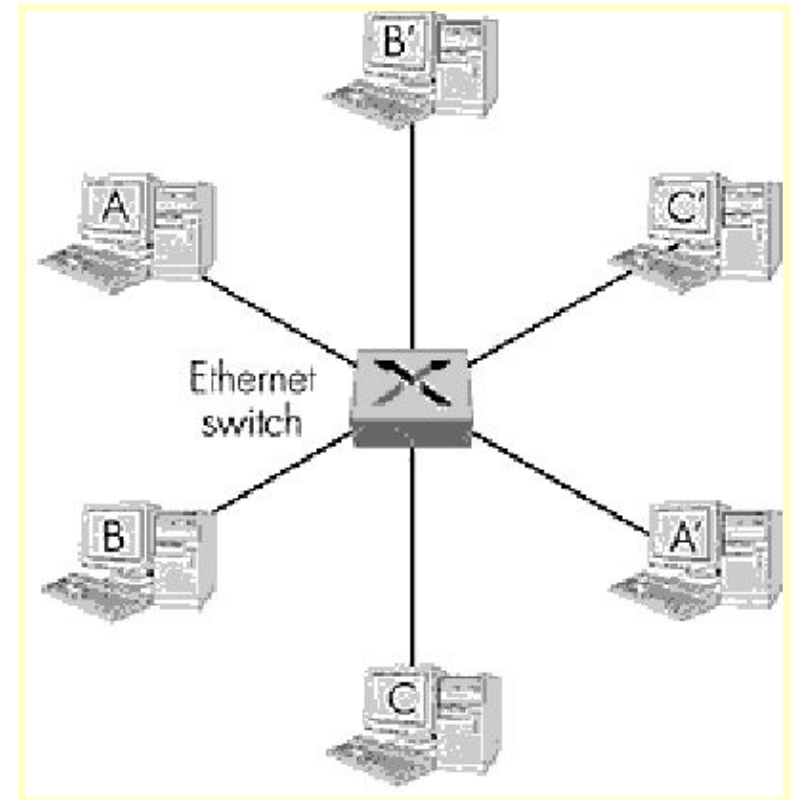
Routers vs. Bridges - 2

Routers + and -

- + arbitrary topologies can be supported, cycling is limited by TTL counters (and good routing protocols)**
- + provide firewall protection against broadcast storms**
- require IP address configuration (not plug and play)**
- require higher processing bandwidth**
- Bridges do well in small (few hundred hosts) while routers used in large networks (thousands of hosts)**

Ethernet Switches - 1

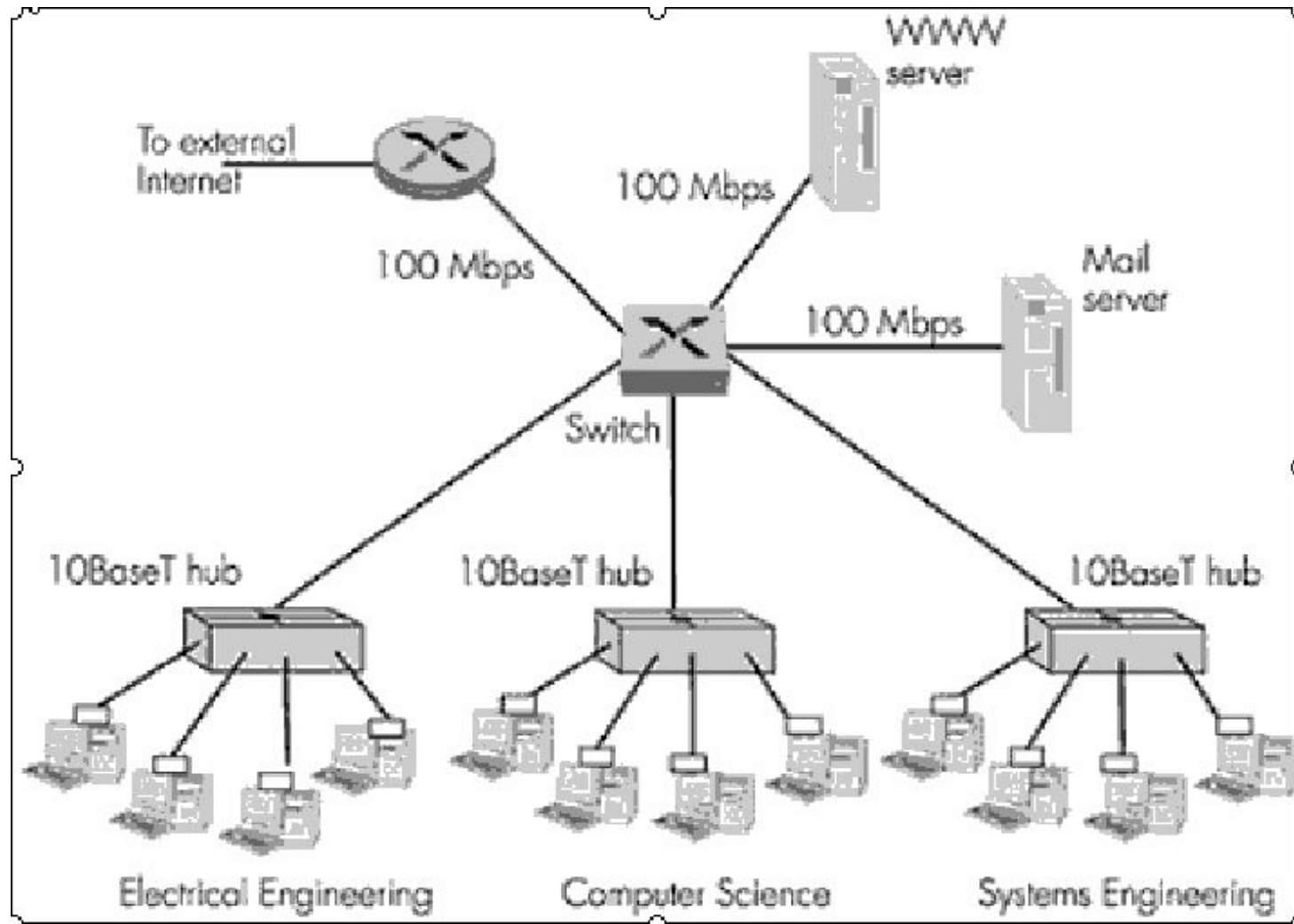
- layer 2 (frame) forwarding, filtering using LAN addresses
- **Switching:** A-to-B and A'-to-B' simultaneously, no collisions
- large number of interfaces
- often: individual hosts, star-connected into switch
 - Ethernet, but no collisions!



Ethernet Switches - 2

- **cut-through switching:** frame forwarded from input to output port without awaiting for assembly of entire frame
 - slight reduction in latency
- combinations of shared/dedicated, 10/100/1000 Mbps interfaces

Ethernet Switches - 3



Five Categories of Network Devices

